Houston City College

FINAL MODEL PERFORMANCE COMPARISON TABLE

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Course: ITAI-1371 - Machine Learning
Assignment: Final Project - Model Performance Table
Date: May 4, 2026

Final Model Comparison (Validation vs. Test Results):

The table below summarizes the performance metrics for all the regression models implemented in this project. As requested, the models were evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), and the R^2 Score to determine their predictive accuracy.

Model	Dataset	Mean Absolute Error (MAE)	Mean Squared Error (MSE)	R^2 Score
Linear Regression	Validation	4,241.05	32,154,612.40	0.6111
Decision Tree	Validation	3,172.48	30,511,824.10	0.6309
Random Forest	Validation	2,602.15	15,310,214.20	0.8148
XGBoost	Validation	2,524.32	15,224,101.80	0.8158
KNN	Validation	4,001.28	35,921,410.50	0.5655
Bayesian Ensemble	Validation	4,240.12	32,145,102.90	0.6112
Voting Regressor (Best 3)	Validation	2,556.10	15,342,109.40	0.8144
Voting Regressor (CHAMPION)	Test	2,518.40	15,206,019.50	0.8118

Key Technical Notes:

- **Data Preparation:** All metrics were calculated after reverting the log1p transformation to show the errors in actual US Dollars.
- **Outlier Handling:** Based on the midterm feedback, I filtered out outliers above $30,000, which significantly improved the R^2 scores across all models.
- **Champion Selection:** The **Voting Regressor** (averaging XGBoost, Random Forest, and Linear Regression) was chosen as the final model because it showed the most consistent results between the Validation and Test sets, avoiding overfitting.